

Sorting of Data

11 June 2026 18:38

A →

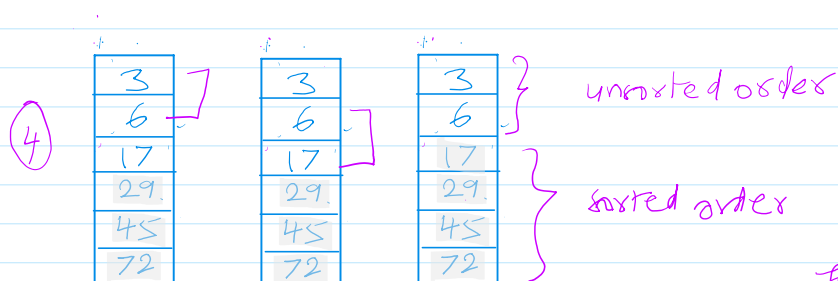
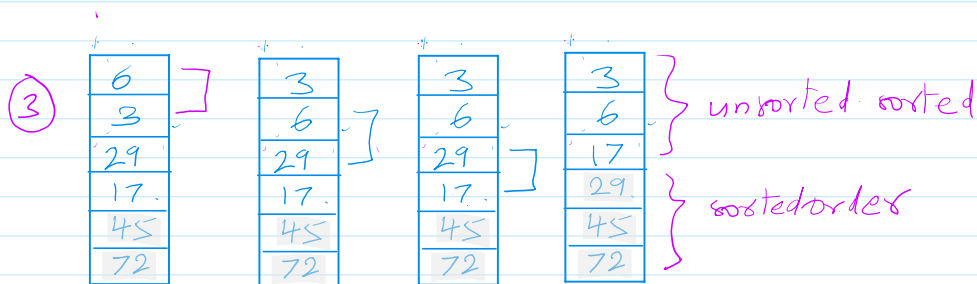
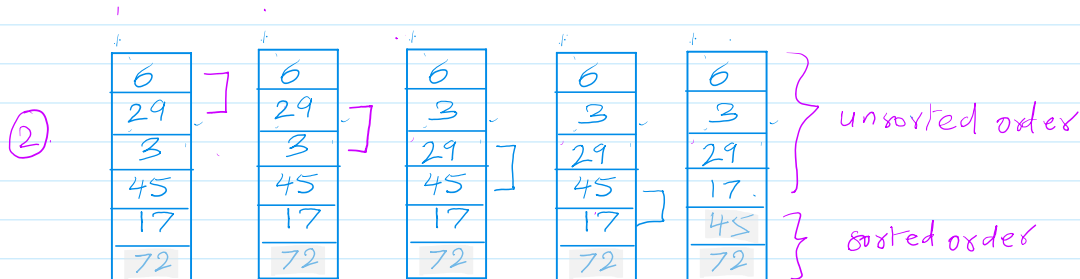
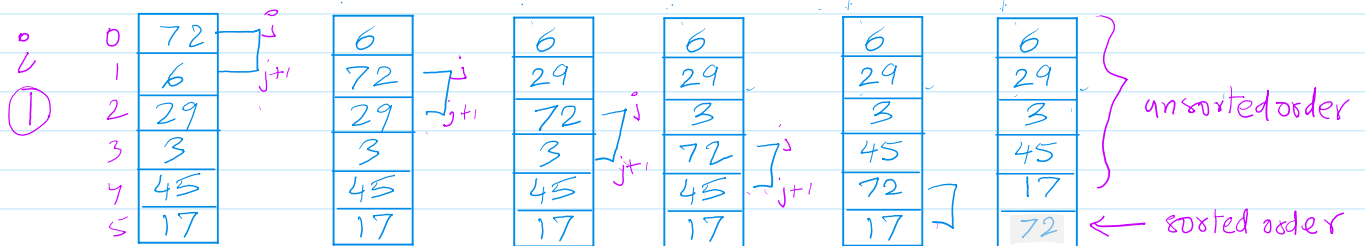
72	6	29	3	45	17	2	9	40	32
----	---	----	---	----	----	---	---	----	----

Arrange the elements in ascending order.

Fundamental sorting techniques are of 3 types:

1. Bubble Sort
2. Selection Sort
3. Insert Sort

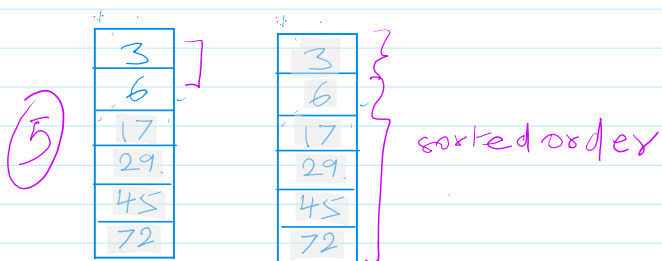
1. Bubble Sort: In bubble sort we compare adjacent elements and swap them, depending upon a condition. For ascending, the greater element and smaller element will change its position.



for n element we need n-1 steps

2 7

Total number of comparison



Total number of comparison
For 6 element, $5+4+3+2+1=15$

```
package MyArrays;
import java.util.*;
```

```
/**
 * Write a description of class ArrayUtility here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class ArrayUtility
{
    public static void printArray(int A[], String msg){
        System.out.print("\n" + msg + "\n[");
        int i;
        for(i = 0; i < A.length - 1; i++){
            System.out.print(A[i]+", ");
        }
        System.out.print(A[i] + "]\n");
    }

    public static void inputArray(int A[], String msg){
        System.out.print("\n" + msg + "\n");
        int i;
        Scanner sc = new Scanner(System.in);
        for(i = 0; i < A.length; i++){
            System.out.print("[ "+i+"]: ");
            A[i] = sc.nextInt();
        }
    }
}
```

```
package MyArrays;
```

```
/**
 * Write a description of class SortingAlgos here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class SortingAlgorithms
{
    //Sorting array in Ascending order
```

```

public static void bubbleSort(int A[]){
    int i, j, t;
    for(i = 0; i < A.length - 1; i++){
        for(j = 0; j < A.length - 1 - i; j++){
            if(A[j] > A[j+1]){
                t = A[j];
                A[j] = A[j+1];
                A[j+1] = t;
            }
        }
    }
}
}
}
}

```

```

package MyArrays;
import java.util.*;

```

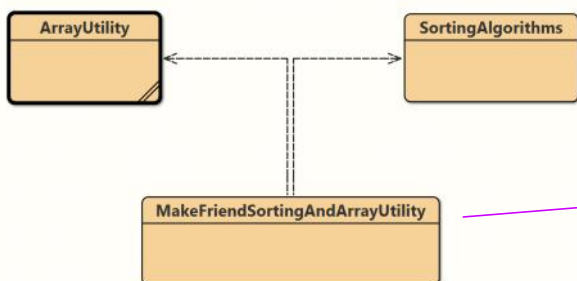
```

/**
 * Write a description of class MakeFriendSortingAndArrayUtility here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class MakeFriendSortingAndArrayUtility
{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        int A[], size;
        System.out.print("\nEnter number of elements: ");
        size = sc.nextInt();
        A = new int[size];
        ArrayUtility.inputArray(A, "Enter " + size + " elements in A[]: ");
        ArrayUtility.printArray(A, "Array A[] before sorting: ");
        SortingAlgorithms.bubbleSort(A);
        ArrayUtility.printArray(A, "Array A[] after sorting: ");
    }
}

```

↑
class reference, because printArray() method is static in ArrayUtility class.

Class reference is used to call static methods



→ Functions present in ArrayUtility and SortingAlgorithms are called in MakeFriendSortingAndArrayUtility class.